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CMPUT 365: Introduction to Reinforcement Learning

Worksheet 7: Temporal Difference Learning Methods for Control

Question 1. Consider the following MDP, with three states B, C and D ($\mathcal{S} = \{B, C, D\}$), and 2 actions ($\mathcal{A} = \{1, 2\}$), with $\gamma = 1.0$. Assume the action values are initialized $Q(s, a) = 0 \ \forall \ s \in \mathcal{S}$ and $a \in \mathcal{A}$. The agent takes actions according to an ϵ-greedy policy with $\epsilon = 0.1$.

(a) What is the optimal policy for this MDP and what are the action-values corresponding to the optimal policy: $q^*(s, a)$?

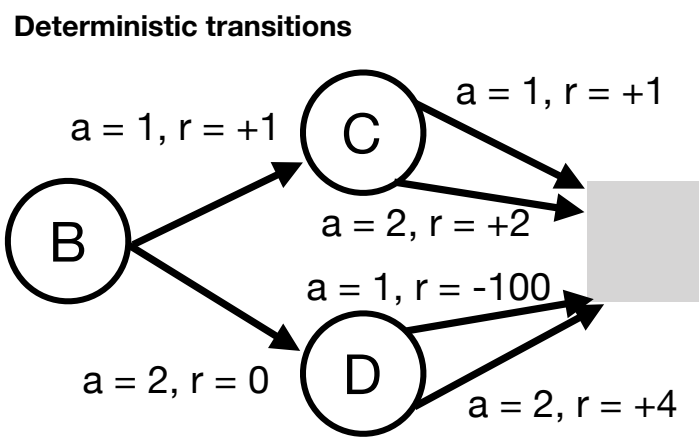
(b) Imagine the agent experienced a single episode, and the following experience: $S_0 = B, A_0 = 2, R_1 = 0, S_1 = D, A_1 = 2, R_2 = 4$. What are the Sarsa updates during this episode, assuming $\alpha = 0.1$? Start with state B, and perform the Sarsa update, then update the value of state D.

(c) Using the sample episode above, compute the updates Q-learning would make, with $\alpha = 0.1$. Again, start with state B, and then state D.

(d) Let's consider one more episode: $S_0 = B, A_0 = 2, R_1 = 0, S_1 = D, A_1 = 1, R_2 = -100$. What would the Sarsa updates be? And what would the Q-learning updates be?

(e) Assume you see one more episode, and it's the same as in (d). Once more, update the action values for Sarsa and Q-learning. What do you notice?

(f) What policy does Q-learning converge to? What policy does Sarsa converge to?



(a) Using the Bellman equation for the optimal action value function we obtain (recall that $q_*(s, a) = \sum_{s',r} p(s', r|s, a)[r + \gamma \max_{a'} q_*(s', a')]$):

$q_*(C, 1) = 1,$ $\quad q_*(C, 2) = 2,$
$q_*(D, 1) = -100,$ $\quad q_*(D, 2) = 4,$
$q_*(B, 1) = 1 + q_*(C, 2) = 3,$ $\quad q_*(B, 2) = 0 + q_*(D, 2) = 4.$

The optimal policy is greedy w.r.t. this value function. So, $\pi_*(2|C) = 1, \pi_*(2|D) = 1, \pi_*(2|B) = 1$, and zero for all other state action pairs.

(b) **SARSA:**

$Q(B, 2) = Q(B, 2) + \alpha \cdot [r(B, 2) + \gamma Q(D, 2) - Q(B, 2)] = 0 + 0.1 \cdot [0 + 0 - 0] = 0$
$Q(D, 2) = Q(D, 2) + \alpha \cdot [r(D, 2) + \gamma Q(T, \cdot) - Q(D, 2)] = 0 + 0.1 \cdot [4 + 0 - 0] = 0.4$

(c) **Q–learning:**

$Q(B,2) = Q(B,2) + \alpha \cdot [r(B,2) + \gamma \max_{a'} Q(D,a') - Q(B,2)] = 0 + 0.1 \cdot [0 + 0 - 0] = 0$

$Q(D,2) = Q(D,2) + \alpha \cdot [r(D,2) + \gamma \max_{a'} Q(T,a') - Q(D,2)] = 0 + 0.1 \cdot [4 + 0 - 0] = 0.4$

(d) **SARSA:**

$Q(B,2) = Q(B,2) + \alpha \cdot [r(B,2) + \gamma Q(D,1) - Q(B,2)] = 0 + 0.1 \cdot [0 + 0 - 0] = 0$

$Q(D,1) = Q(D,1) + \alpha \cdot [r(D,1) + \gamma Q(T,\cdot) - Q(D,1)] = 0 + 0.1 \cdot [-100 + 0 - 0] = -10$

Q–learning:

$Q(B,2) = Q(B,2) + \alpha \cdot [r(B,2) + \gamma \max_{a'} Q(D,a') - Q(B,2)] = 0 + 0.1 \cdot [0 + 0.4 - 0] = 0.04$

$Q(D,1) = Q(D,1) + \alpha \cdot [r(D,1) + \gamma \max_{a'} Q(T,a') - Q(D,1)] = 0 + 0.1 \cdot [-100 + 0 - 0] = -10$

(e) **SARSA:**

$Q(B,2) = Q(B,2) + \alpha \cdot [r(B,2) + \gamma Q(D,1) - Q(B,2)] = 0 + 0.1[0 + -10 - 0] = -1$, and

$Q(D,1) = Q(D,1) + \alpha \cdot [r(D,1) + \gamma Q(T,\cdot) - Q(D,1)] = -10 + 0.1[-100 + 0 + 10] = -19.$

Q–learning:

$Q(B,2) = Q(B,2) + \alpha \cdot [r(B,2) + \gamma \max_{a'} Q(D,a') - Q(B,2)] = 0.04 + 0.1 \cdot [0 + 0.4 - 0.04] = 0.076$

$Q(D,1) = Q(D,1) + \alpha \cdot [r(D,1) + \gamma \max_{a'} Q(T,a') - Q(D,1)] = -10 + 0.1 \cdot [-100 + 0 + 10] = -19$

We observe that the SARSA and Q–learning updates are the same for $Q(D,1)$, and they are different for $Q(B,2)$. For $Q(B,2)$, Q–learning makes the update using the next state action pair as $Q(D,2)$, even though the sampled next-state-action pair is $(D,1)$. Because it has a higher value than the other action value $Q(D,1)$.

(f) Q–learning converges to the optimal greedy policy. Sarsa converges to the optimal policy in behavior policy class (for example the optimal ϵ–greedy policy). The Cliff Grid World illustrates this.

Question 2. In Monte Carlo control, we required that every state-action pair be visited infinitely often. One way this can be guaranteed is by using exploring starts. Can we use exploring starts for Sarsa? Further, we have talked about using Sarsa with an ϵ-greedy policy. Can we use Monte Carlo with an ϵ-greedy policy? Does this ensure sufficient exploration?

Yes. From the textbook "Sarsa converges with probability 1 to an optimal policy and action-value function as long as all state–action pairs are visited an infinite number of times". And since exploring starts explicitly satisfies this assumption, we can use Sarsa with exploring starts without any issues.

Yes, Monte–Carlo can be used with an ϵ–greedy policy. An ϵ–greedy policy is also an ϵ–soft policy. And we know that the policy iteration theorem holds for ϵ–soft policy (see the textbook, Chapter 5); therefore, MC works with an ϵ–greedy policy. Yes, this ensures sufficient exploration in the limit, since each action has a non–zero probability of being chosen.

Question 3. Exercise 6.11 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

Question 4. Exercise 6.12 from Sutton & Barto (2018).

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Question 5. Exercise 6.13 from Sutton & Barto (2018).

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